

Order	Team Name	Function	Days Needed	Deliverables	Handoff	Further Steps Ahead	Tools
1	Intent Agent Team	Collect academic data, clean it, build pipelines, create embeddings	3 (Day 1–3)	Raw CSV/JSON data, chunked text files, embeddings in PostgreSQL	Pass structured data (even partial) to Retriever Agent	Ensure embeddings are query-ready	Python (Pandas), LangChain, Hugging Face, PostgreSQL
2	Retriever Agent Team	Implement retrieval system for RAG (vector search, indexing)	3 (Day 2–4)	FAISS/Weaviate index, retrieval API, query-document mapping	Provide retrieval API to Query Generator	Optimize retrieval latency	FAISS / Weaviate / Pinecone
3	Table Agent Team	Structure tabular academic data for efficient querying	2 (Day 2–3)	Normalized tables, schema definitions, SQL-ready datasets	Pass schema to Column Pruning Agent	Ensure schema aligns with retriever	PostgreSQL, SQLAlchemy
4	Column Pruning Agent Team	Optimize database schema by pruning irrelevant columns	2 (Day 4–5)	Optimized schema, reduced storage footprint, documentation	Provide optimized schema to Query Generator	Maintain schema evolution logs	dbt, Apache Arrow
5	Query Generator Agent Team	Generate SQL/semantic queries from user prompts	3 (Day 5–7)	Query generation module, test cases, integration with retriever	Pass queries to SQL Validator	Improve semantic-to-SQL accuracy	LangChain SQLChain, OpenAI API
6	SQL Validator Agent Team	Validate generated queries for correctness and safety	2 (Day 6–7)	Query validation engine, error handling, safe execution logs	Pass validated queries to Retriever + Synthesis Agent	Add guardrails for unsafe queries	Python (sqlparse, SQLFluff)
7	Web Development Team	Build front-end interface for chatbot (UI/UX, dashboards)	4 (Day 5–8)	Web app (React/Next.js), chatbot UI, API integration hooks	Integrate with Synthesis Agent outputs	Enhance UI with feedback loop	React / Next.js, TailwindCSS, FastAPI
8	DevOps Team	Set up CI/CD, containerization, deployment pipelines	4 (Day 6–9)	Docker/Kubernetes setup, CI/CD workflows, monitoring dashboards	Deploy web app + backend services	Scale infra, add observability	Docker, Kubernetes, GitHub Actions, Prometheus/Grafana
9	Synthesis Agent Team	Combine retrieved chunks into coherent responses	3 (Day 8–10)	Response synthesis module, summarization engine, test outputs	Provide synthesized responses to Audit & Feedback Agent	Improve reasoning + coherence	Hugging Face Transformers, LangChain
10	Audit & Feedback Agent Team	Evaluate chatbot responses, collect feedback, fine-tune	3 (Day 11–13)	Evaluation reports, feedback loop integration, retraining scripts	Feed insights back to Intent + Retriever teams	Continuous improvement cycle	MLflow, EvidentlyAI
—	Integration & Finalization	End-to-end testing, bug fixes, performance tuning	5 (Day 14–18)	Stable integrated system, optimized latency, final demo	—	Production readiness, monitoring	Postman, Grafana, Logging tools